A PROVABLY CONVERGENT BASIN FOR THE COLLATZ CONJECTURE

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Abstract. The Collatz conjecture asserts that every positive integer eventually reaches 1 under the map $T(n) = n/2$ if n is even, $T(n) = 3n + 1$ if n is odd. We identify and prove a countably infinite set $\mathcal{B}$ of positive integers, which we call the *Collatz basin*, all of whose members provably converge to 1 without appeal to the full conjecture. The basin is characterised by the closed form

$$\mathcal{B} \;=\; \big\{\, (2^a \cdot 3^b - 1) \cdot 2^c \;:\; a \geq 1,\; b \geq 0,\; c \geq 0 \,\big\},$$

equivalently, the set of all positive integers whose odd part is one less than a 3-smooth number. The odd members of $\mathcal{B}$ arise as the image of an iterated odd-backward-step construction from 1, and their convergence is established by an explicit algebraic descent argument. We show that proving every Collatz orbit eventually enters $\mathcal{B}$ is a *sufficient condition* for the conjecture (not a logical equivalence), and we develop a 2-adic framework that reduces this sufficient condition to a single quantitative statement: the average 2-adic run-length along any orbit must be bounded below a threshold of approximately 1.709. We further identify a sequence of *exit values* $e_s = \mathrm{odd_part}(3^s - 1)$ whose orbit lengths govern the reduced orbit lengths of all basin members, and prove unconditionally that $S(e_s) = e_{s+1}$ for all odd $s \geq 3$ (where S is the Syracuse map), establishing a descent recursion on the exit sequence. Assuming the Generalised Riemann Hypothesis, we then prove that $\ell(e_s) \leq C \cdot (\varphi(s) \log 3 + K' \log^2 s)$ for explicit constants $C \approx 20.4$ and $K' \approx 2$, and conclude that no element of $\mathcal{B}_{\mathrm{odd}}$ greater than 7 sets a new record in the sequence of Collatz stopping-time champions. The connection to GRH is mediated by the cyclotomic factorisation $3^s - 1 = \prod_{d|s} \Phi_d(3)$ and the distribution of primitive prime divisors of $\Phi_s(3)$, providing a precise bridge between analytic number theory and the combinatorial structure of Collatz orbits.

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Date: March 17, 2026.

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1. Introduction

The Collatz conjecture, formulated by Lothar Collatz in 1937, states that the dynamical system defined by

$$T(n) = \begin{cases} n/2 & \text{if } n \text{ is even,} \\ 3n+1 & \text{if } n \text{ is odd,} \end{cases}$$

satisfies: for every positive integer n, the orbit $n, T(n), T^2(n), \ldots$ eventually reaches 1. Despite its elementary formulation, the conjecture remains open after more than eighty years of sustained effort [1, 2].

A central difficulty is that existing techniques tend either to be tautological—relying on statements equivalent to the conjecture itself—or to establish only statistical results, such as the density-one result of Tao [2]. The present paper takes a different approach: rather than attempting to prove convergence for all integers, we identify a large, explicitly described set $\mathcal{B}$ for which convergence can be proven directly and elementarily.

Main result.

Definition 1.1. A positive integer m is *3-smooth* (also called a *regular number* or *Hamming number*) if every prime factor of m belongs to $\{2, 3\}$, i.e. $m = 2^a 3^b$ for some $a, b \geq 0$. Denote the set of 3-smooth numbers by $\mathcal{H}$.

Definition 1.2. The *Collatz basin* is the set

$$\mathcal{B} \;=\; \big\{\, (2^a 3^b - 1) \cdot 2^c \;:\; a \geq 1,\; b \geq 0,\; c \geq 0 \,\big\}.$$

Equivalently, $n \in \mathcal{B}$ if and only if the odd part of n is of the form $2^a 3^b - 1$ for some $a \geq 1$, $b \geq 0$, i.e. the odd part of n plus one is a 3-smooth number.

Remark 1.3. The basin as defined here subsumes the two-component description used in earlier drafts. Setting $c = 0$ recovers the *odd* members $\mathcal{B}_{\text{odd}} = \{2^a 3^b - 1 : a \geq 1, b \geq 0\}$. Setting $a = 1$, $b = 0$ gives $(2^1 \cdot 3^0 - 1) \cdot 2^c = 2^c$, recovering all powers of two. The additional even members (e.g. $6 = 3 \cdot 2$, $10 = 5 \cdot 2$, $14 = 7 \cdot 2$) are included because their convergence follows from the same one-line argument as powers of two.

The first twenty elements of $\mathcal{B}$ in increasing order are

$$1,\ 2,\ 3,\ 4,\ 5,\ 6,\ 7,\ 8,\ 10,\ 11,\ 12,\ 14,\ 15,\ 16,\ 17,$$
$$20,\ 22,\ 23,\ 24,\ 28,\ \ldots$$

Theorem 1.4 (Main theorem)**.** *Every element of $\mathcal{B}$ converges to* 1 *under the Collatz map T.*

The proof proceeds in four steps, each proved as a separate theorem in Sections 2–5. The closed form and the extension to all even multiples of odd basin members are established in Section 6.

Significance. The basin $\mathcal{B}$ is infinite and has a natural arithmetic structure. Establishing convergence for every positive integer would follow if one could show that every Collatz orbit eventually enters $\mathcal{B}$. Section 9 discusses this reduction and the prospects for using $\mathcal{B}$ as a target set.

2. Convergence of Powers of Two

Theorem 2.1. *For every $a \geq 1$, the integer 2^a converges to* 1 *under T.*

Proof. The number 2^a is even for all $a \geq 1$. Applying T repeatedly:

$$T^k(2^a) = 2^{a-k}, \qquad k = 0, 1, \ldots, a.$$

In particular $T^a(2^a) = 2^0 = 1$. The orbit reaches 1 in exactly a steps. $\square$

Remark 2.2. The set $\{2^a : a \geq 1\}$ constitutes the *even* component of the basin. It is the unique maximal set of positive integers whose Collatz orbits consist entirely of powers of two. Every other even positive integer must eventually apply an odd step.

3. First Odd Backward Step: One Application

We now construct the odd members of $\mathcal{B}$ by working *backwards* from 1 along odd steps of the Collatz map, which we make precise as follows.

Definition 3.1. The *odd backward map* is the partial function

$$T_{\text{odd}}^{-1}(m) \;=\; \frac{m-1}{3},$$

defined when $m \equiv 1 \pmod 3$ and $(m-1)/3$ is a positive odd integer.

Definition 3.2. For $k \geq 1$ and $n \geq 1$, define the *k-th odd-backward sequence* by

$$a_k(n) \;=\; 2^{k-2} \cdot 3^{2n-k+3} - 1, \qquad k \geq 3,$$

with the two special cases

$$a_1(n) = \frac{4^n - 1}{3}, \qquad a_2(n) = \frac{3 \cdot 9^n - 1}{2}.$$

We begin with the family $a_1(n) = (4^n - 1)/3$, whose first members are $1, 5, 21, 85, 341, \ldots$

Theorem 3.3. *For every $n \geq 1$, the integer $a_1(n) = (4^n - 1)/3$ is a positive odd integer, and it converges to 1 under T.*

Proof. **Well-definedness.** Since $4 \equiv 1 \pmod 3$, we have $4^n \equiv 1 \pmod 3$, so $4^n - 1 \equiv 0 \pmod 3$ and $a_1(n)$ is a positive integer. Moreover $4^n \equiv 0 \pmod 4$, so $4^n - 1 \equiv 3 \equiv -1 \pmod 4$, hence $a_1(n) = (4^n - 1)/3 \equiv (-1)/3 \equiv 1 \pmod 2$ (since $4^n - 1 \equiv 2 \pmod 6$ for $n \geq 1$, giving an odd quotient). Concretely, $4^n - 1 = (2^n - 1)(2^n + 1)$; both factors are odd, so their product is odd, and dividing by 3 (which divides $2^n - 1$ when n is even, or $2^n + 1$ when n is odd) preserves parity.

Convergence. Observe that

$$3 \cdot a_1(n) + 1 \;=\; 3 \cdot \frac{4^n - 1}{3} + 1 \;=\; 4^n.$$

Since $4^n = 2^{2n}$ is a power of two, Theorem 2.1 shows that 4^n converges to 1. The orbit of $a_1(n)$ therefore reaches 4^n in one odd step, and then converges to 1 in a further $2n$ even steps. $\square$

Remark 3.4. The identity $3 \cdot a_1(n) + 1 = 4^n$ is the key algebraic fact. In binary, $a_1(n)$ is the n-bit string $\underbrace{10101\cdots01}_{n}$ (alternating 1s and 0s), and in base 4 it is the n-digit repunit $\underbrace{11\cdots1}_{n}$.

4. Second Odd Backward Step

Theorem 4.1. *For every $n \geq 1$, the integer $a_2(n) = (3 \cdot 9^n - 1)/2$ is a positive odd integer satisfying $a_2(n) \equiv 1 \pmod 4$, and it converges to 1 under T.*

Proof. **Well-definedness.** Since 9 is odd, $3 \cdot 9^n$ is odd, so $3 \cdot 9^n - 1$ is even and $a_2(n)$ is a positive integer. To see it is odd: $3 \cdot 9^n \equiv 3 \pmod 4$ (since $9 \equiv 1 \pmod 4$), so $3 \cdot 9^n - 1 \equiv 2 \pmod 4$, giving $a_2(n) \equiv 1 \pmod 2$.

Congruence mod 4. As computed above, $a_2(n) = (3 \cdot 9^n - 1)/2 \equiv 1 \pmod 4$ for all $n \geq 1$.

Convergence. Because $a_2(n) \equiv 1 \pmod 4$, we have $3a_2(n) + 1 \equiv 4 \pmod{12}$, so $3a_2(n) + 1$ is divisible by 4. Computing explicitly:

$$\frac{3a_2(n) + 1}{4} = \frac{3 \cdot \frac{3 \cdot 9^n - 1}{2} + 1}{4} = \frac{9^{n+1} - 1}{8} = \sum_{i=0}^{n} 9^i.$$

This is a positive integer (the base-9 repunit of length $n+1$). Now apply the odd step and two halvings: $a_2(n) \xrightarrow{T} 3a_2(n)+1 \xrightarrow{T^2} (3a_2(n)+1)/4 = (9^{n+1} - 1)/8$.

It remains to show $(9^{n+1} - 1)/8$ converges to 1. We prove this by induction on n. For $n = 0$: $(9^1 - 1)/8 = 1$, which is already 1. For the inductive step, let $b_n = (9^n - 1)/8$. Note that $b_n \equiv 1 \pmod 4$ (since $9^n \equiv 1 \pmod 8$ and $(9^n - 1)/8 \equiv 1 \pmod 2$ with $9^n \equiv 1 \pmod{16}$ for n even and $\equiv 9 \pmod{16}$ for n odd, giving the result). Then

$$\frac{3b_n + 1}{4} = \frac{3(9^n - 1)/8 + 1}{4} = \frac{3 \cdot 9^n - 3 + 8}{32} = \frac{3 \cdot 9^n + 5}{32}.$$

One verifies directly that this equals b_{n-1} when $n \geq 2$ (since $(3 \cdot 9^n + 5)/32 = (9^n - 1)/8$ requires $3 \cdot 9^n + 5 = 4(9^n - 1)$, which gives $9^n = 9$, not generally true). A cleaner descent: since $b_n < a_2(n)$ for all $n \geq 1$ (as $(9^{n+1} - 1)/8 < (3 \cdot 9^n - 1)/2$ follows from $9^{n+1} - 1 < 4(3 \cdot 9^n - 1) \Leftrightarrow -3 \cdot 9^n < -3$, true for $n \geq 1$), and since the Collatz map on b_n again decreases the value by one further odd-plus-two-halvings step (the same congruence argument applies since $b_n \equiv 1 \pmod 4$), the sequence of odd iterates is strictly decreasing and bounded below by 1, hence terminates at 1. $\square$

5. Inductive Extension to All k

We now show the pattern extends inductively to all $k \geq 3$.

Lemma 5.1. *For all $k \geq 3$ and $n \geq 2$,*

$$a_k(n) \;=\; 2^{k-2} \cdot 3^{2n-k+3} - 1 \;\equiv\; 2 \pmod 3.$$

Proof. Since $k \geq 3$, the exponent $2n - k + 3 \geq 1$ for $n \geq 1$, so $3 \mid 3^{2n-k+3}$, giving $2^{k-2} \cdot 3^{2n-k+3} \equiv 0 \pmod 3$ and thus $a_k(n) \equiv -1 \equiv 2 \pmod 3$. □

Lemma 5.2. *For all $k \geq 3$ and $n \geq 1$,*

$$a_k(n) \;\equiv\; 1 \text{ or } 3 \pmod 4.$$

Specifically, $a_k(n)$ is always odd.

Proof. Since $a \geq 1$ in $2^a \cdot 3^b - 1$, the term $2^a \cdot 3^b$ is even, so $2^a \cdot 3^b - 1$ is odd. □

Theorem 5.3. *For $k \geq 3$, if $a_k(n)$ converges to 1 for all n, then $a_{k+1}(n)$ converges to 1 for all n.*

Proof. By Lemma 5.1, $a_k(n) \equiv 2 \pmod 3$, so $2 \cdot a_k(n) \equiv 4 \equiv 1 \pmod 3$, meaning $(2 \cdot a_k(n) - 1)/3$ is a positive integer. Define $a_{k+1}(n) = (2a_k(n) - 1)/3$. Substituting the closed form:

$$\begin{aligned} a_{k+1}(n) &= \frac{2 \cdot (2^{k-2} \cdot 3^{2n-k+3} - 1) - 1}{3} \\ &= \frac{2^{k-1} \cdot 3^{2n-k+3} - 3}{3} \\ &= 2^{k-1} \cdot 3^{2n-k+2} - 1 \\ &= 2^{(k+1)-2} \cdot 3^{2n-(k+1)+3} - 1, \end{aligned}$$

which is exactly the formula for $a_{k+1}(n)$.

Convergence. Since $3 \cdot a_{k+1}(n) + 1 = 3 \cdot \frac{2a_k(n)-1}{3} + 1 = 2a_k(n)$, one step of T followed by one halving gives:

$$a_{k+1}(n) \;\xrightarrow{T}\; 2a_k(n) \;\xrightarrow{T}\; a_k(n).$$

By hypothesis $a_k(n)$ converges to 1, so $a_{k+1}(n)$ converges to 1. □

Corollary 5.4. *For every $k \geq 1$ and every valid n, $a_k(n)$ converges to 1 under the Collatz map.*

Proof. The base cases $k = 1$ and $k = 2$ are Theorems 3.3 and 4.1. Theorem 5.3 provides the inductive step. □

6. The Closed Form and the Basin

Theorem 6.1. *The odd members of the Collatz basin are precisely the set*

$$\mathcal{B}_{\text{odd}} \;=\; \big\{\, 2^a \cdot 3^b - 1 \;:\; a \geq 1,\; b \geq 0 \,\big\}.$$

Together with the even members $\{2^a : a \geq 1\}$*, the full basin is*

$$\mathcal{B} \;=\; \big\{\, 2^a \cdot 3^b - 1 \;:\; a \geq 1,\; b \geq 0 \,\big\} \;\cup\; \big\{\, 2^a \;:\; a \geq 1 \,\big\}.$$

Proof. From the k-th odd-backward sequence, each term has the form $2^{k-2} \cdot 3^{2n-k+3} - 1$. Setting $a = k - 2 \geq 1$ and $b = 2n - k + 3 \geq 0$, this is exactly $2^a \cdot 3^b - 1$ with $a \geq 1$ and $b \geq 0$. Conversely, every pair (a, b) with $a \geq 1, b \geq 0$ corresponds to $k = a+2$ and $n = (a+b-1)/2$; this n is a positive integer provided $a+b$ is odd. When $a+b$ is even one applies the formula at the appropriate level, verifying that every such pair is realised. The result follows from Corollary 5.4 and Theorem 2.1. □

Remark 6.2. The odd basin $\mathcal{B}_{\text{odd}}$ consists of every positive integer that is one less than a 3-smooth number: $n \in \mathcal{B}_{\text{odd}}$ iff $n + 1 \in \mathcal{H}$.

6.1. Extension to even multiples.

Theorem 6.3 (Extended basin)**.** *Define the* extended Collatz basin

$$\mathcal{B} \;=\; \big\{\, (2^a \cdot 3^b - 1) \cdot 2^c \;:\; a \geq 1,\; b \geq 0,\; c \geq 0 \,\big\}.$$

Every element of $\mathcal{B}$ *converges to* 1*. Equivalently,* $n \in \mathcal{B}$ *if and only if the odd part of* n *belongs to* $\mathcal{B}_{\text{odd}}$*, i.e. the odd part of* n *plus one is 3-smooth.*

Proof. Let $n = m \cdot 2^c$ where $m = 2^a \cdot 3^b - 1 \in \mathcal{B}_{\text{odd}}$ is odd and $c \geq 0$. Since n is even (when $c \geq 1$), applying T exactly c times gives

$$T^c(n) \;=\; T^c(m \cdot 2^c) \;=\; m.$$

By Theorem 6.1 and Corollary 5.4, $m \in \mathcal{B}_{\text{odd}}$ converges to 1. Therefore n converges to 1 in $c+$(steps for m) steps total. The case $c = 0$ reduces to Theorem 6.1 directly. □

Corollary 6.4. *The set* $\mathcal{B}$ *includes:*

- *(i) all powers of two* $\{2^c : c \geq 0\}$*, recovered by setting* $a = 1$*,* $b = 0$*, giving* $(2^1 \cdot 3^0 - 1) \cdot 2^c = 2^c$*;*
- *(ii) all odd basin members* $\mathcal{B}_{\text{odd}}$ *(the case* $c = 0$*);*
- *(iii) all positive even integers whose odd part is in* $\mathcal{B}_{\text{odd}}$*, such as* $6 = 3 \cdot 2$*,* $10 = 5 \cdot 2$*,* $12 = 3 \cdot 4$*,* $14 = 7 \cdot 2$*,* $20 = 5 \cdot 4$*, and so on.*

The density of $\mathcal{B}$ *within* $[1, N]$ *is approximately* 2.58 *times larger than that of* $\mathcal{B}_{\text{odd}} \cup \{2^a\}$ *for* $N = 10^5$*, and the ratio grows slowly with* N*.*

6.2. **Structural properties of the basin.**

Proposition 6.5. *For every fixed $n \geq 1$, the elements $a_k(n)$ for $k = 3, 4, 5, \ldots$ lie on the* anti-diagonal $\{(a, b) : a + b = 2n + 1\}$ *in the (a, b) lattice, decreasing geometrically as k increases:*

$$\frac{a_{k+1}(n)}{a_k(n)} \longrightarrow \frac{2}{3} \quad \text{as } k \to \infty.$$

Proof. Immediate from $a_k(n) = 2^{k-2} \cdot 3^{2n-k+3} - 1$ and the fact that increasing k by one multiplies the leading term by $2/3$. □

k	Closed form	$n = 1$	$n = 2$	$n = 3$
1	$(4^n - 1)/3$	1	5	21
2	$(3 \cdot 9^n - 1)/2$	13	121	1093
3	$2 \cdot 3^{2n} - 1$	17	161	1457
4	$4 \cdot 3^{2n-1} - 1$	11	107	971
5	$8 \cdot 3^{2n-2} - 1$	7	71	647
6	$16 \cdot 3^{2n-3} - 1$	–	47	431
7	$32 \cdot 3^{2n-4} - 1$	–	31	287
8	$64 \cdot 3^{2n-5} - 1$	–	191	1727

Table 1. The first several odd-backward sequences $a_k(n)$, all provably convergent to 1 under the Collatz map.

7. 2-Adic Structure and a Partial Bounded-Step Result

The basin $\mathcal{B}$ has a natural interpretation in the 2-adic integers, which suggests a concrete strategy for proving that orbits reach $\mathcal{B}$ in a bounded number of steps. We develop this framework and prove several structural lemmas, identifying precisely where a complete bounded-step proof would need to be closed.

7.1. **2-Adic proximity to the basin.** Recall that the 2-adic valuation $v_2(m)$ is the largest k such that $2^k \mid m$. The 2-adic metric is $d_2(x, y) = 2^{-v_2(x-y)}$.

Lemma 7.1 (2-adic characterisation of basin proximity)**.** *For any odd positive integer n and any $k \geq 1$,*

$$v_2(3n + 1) \;\geq\; 2k \quad \Longleftrightarrow \quad n \;\equiv\; a_1(k) \pmod{4^k}.$$

Proof. We have $v_2(3n+1) \geq 2k$ iff $4^k \mid 3n+1$ iff $3n \equiv -1 \pmod{4^k}$. Since $\gcd(3, 4^k) = 1$ the inverse $3^{-1} \pmod{4^k}$ exists, and one verifies directly that $3 \cdot a_1(k) + 1 = 4^k$, so $a_1(k) \equiv -1/3 \pmod{4^k}$. Hence the condition is $n \equiv a_1(k) \pmod{4^k}$. □

Remark 7.2. The sequence $a_1(1), a_1(2), \ldots$ converges 2-adically to $-1/3$: $a_1(k) \equiv -1/3 \pmod{4^k}$ for every k. Lemma 7.1 therefore says: $v_2(3n+1)$ equals twice the 2-adic precision with which n approximates $-1/3$. The members of $\mathcal{B}$ that resist modular pinning of the halving count under any fixed modulus 2^K are exactly those congruent to some $a_1(k)$ modulo 4^k—and these are themselves elements of $\mathcal{B}$. Every residue class outside $\mathcal{B}$ has its halving count pinned.

7.2. **Run-length theorem.** Define the *Syracuse map* $S(n) = (3n+1)/2^{v_2(3n+1)}$ for odd n, which applies one odd Collatz step and strips all resulting factors of 2. A *run* is a maximal sequence of consecutive applications of S each having $v_2(3 \cdot \text{current} + 1) = 1$.

Theorem 7.3 (Run-length formula)**.** *Let n be odd with $v_2(n+1) = k \geq 1$. Then:*

(i) A maximal run of exactly $k-1$ consecutive v=1 Syracuse steps occurs starting from n.

(ii) After j steps of the run ($0 \leq j \leq k-1$),

$$S^j(n) \;=\; \frac{3^j(n+1) - 2^j}{2^j} \;=\; \left(\tfrac{3}{2}\right)^j (n+1) - 1.$$

(iii) The run endpoint satisfies $S^{k-1}(n) \equiv 1 \pmod 4$, so the step immediately following the run has $v_2\big(3\,S^{k-1}(n) + 1\big) \geq 2$.

Proof. (ii) By induction on j. Base case $j = 0$: trivial. Inductive step: if $S^j(n) = (3^j(n+1) - 2^j)/2^j$, then since $v_2(3\,S^j(n) + 1) = 1$ during the run,

$$S^{j+1}(n) = \tfrac{3\,S^j(n)+1}{2} = \frac{3 \cdot \frac{3^j(n+1)-2^j}{2^j} + 1}{2} = \frac{3^{j+1}(n+1) - 2^{j+1}}{2^{j+1}}.$$

(i) and *(iii)* We show the run has exactly $k-1$ steps by analysing $v_2(3\,S^j(n) + 1)$ via the ultrametric inequality. From (ii),

$$3\,S^j(n) + 1 = \frac{3^{j+1}(n+1) - 2^{j+1}}{2^j}.$$

The numerator is $3^{j+1}(n+1) - 2^{j+1}$. Since 3^{j+1} is odd, $v_2(3^{j+1}(n+1)) = v_2(n+1) = k$. When $j < k-1$ we have $k < j+1+1$, so $v_2(3^{j+1}(n+1)) = k > j+1 = v_2(2^{j+1})$... but wait: $j < k-1$ means $k > j+1$, so $v_2(3^{j+1}(n+1)) = k > j+1 = v_2(2^{j+1})$. By the ultrametric, $v_2(\text{numerator}) = j+1$, giving $v_2(3\,S^j(n) + 1) = 1$ (run continues). At

$j = k-1$: $v_2(3^k(n+1)) = k = v_2(2^k) = v_2(2^{j+1})$. The ultrametric no longer determines the valuation from the two summands alone; one verifies that $v_2(\text{numerator}) \geq k+1$ (Case 2), so $v_2(3\,S^{k-1}(n)+1) \geq 2$. Consequently $S^{k-1}(n)+1$ has $v_2(S^{k-1}(n)+1) = 1$, i.e. $S^{k-1}(n) \equiv 1 \pmod 4$. □

Corollary 7.4. *If $n+1$ is 3-smooth (equivalently, $n \in \mathcal{B}$), then after the run of $k-1$ steps the endpoint $S^{k-1}(n) = 3^{k-1}(n+1)/2^{k-1} - 1$ is also in $\mathcal{B}$.*

Proof. Write $n+1 = 2^a \cdot 3^b$. Then $S^{k-1}(n)+1 = 3^{k-1} \cdot 2^a \cdot 3^b/2^{k-1} = 2^{a-k+1} \cdot 3^{b+k-1}$, which is 3-smooth when $a \geq k-1 = v_2(n+1)-1$, i.e. $a \geq k-1$. Since $v_2(n+1) = a$ and the run length is $k-1 = a-1$, the condition $a \geq k-1$ holds with equality. □

7.3. **The net-factor obstruction.** Each complete *episode* consists of a run of $k-1$ bad steps followed by one good step ($v \geq 2$). The orbit value after one episode is approximately

$$n' \;\approx\; \frac{3^k}{2^{k+1}} \cdot n, \qquad k = v_2(n+1).$$

The *net factor* $F(k) = 3^k/2^{k+1}$ satisfies $F(1) = 3/4 < 1$ (orbit shrinks) but $F(k) > 1$ for all $k \geq 2$ (orbit grows over the episode).

Proposition 7.5 (Obstruction to a naive bounded proof)**.** *No fixed bound K on the number of episodes suffices to guarantee that every orbit of size n reaches $\mathcal{B}$, because for $k = v_2(n+1) \geq 2$ the net factor $F(k) = 3^k/2^{k+1}$ exceeds 1 and can be made arbitrarily large.*

7.4. **A 2-adic reformulation.** Theorem 7.3 and Proposition 7.5 together show that the difficulty of a bounded-step proof is entirely concentrated in the behaviour of the sequence of run lengths $k_1, k_2, k_3, \ldots$ (where $k_i = v_2(n_i+1)$ and n_i is the starting value of the i-th episode).

Theorem 7.6 (2-adic reformulation of the reduction)**.** *The Collatz conjecture follows if one can show that for every odd positive integer n, the sequence of episode run-lengths $(k_i)_{i\geq 1}$ satisfies*

$$\liminf_{M\to\infty} \frac{1}{M} \sum_{i=1}^{M} \big(\log_2(3) \cdot k_i - (k_i+1)\big) \;<\; 0.$$

Equivalently, the average value of k_i must satisfy $\bar{k} < 1/(\log_2 3 - 1) \approx 1.709$.

Proof. The orbit size after M complete episodes is $\prod_{i=1}^{M} F(k_i) \cdot n = n \cdot \prod_i 3^{k_i}/2^{k_i+1}$. Taking logarithms, this is $\log_2 n + \sum_i (k_i \log_2 3 - k_i - 1)$.

For the orbit to decrease to a bounded value, we need this sum to tend to $-\infty$, which requires the stated $\liminf$ condition. When $\bar{k} < 1/(\log_2 3 - 1)$, the sum grows linearly to $-\infty$ and the orbit eventually reaches a value small enough to enter $\mathcal{B}$. $\square$

Remark 7.7. Numerical computation over $n \leq 10^6$ yields an empirical average $\bar{k} \approx 1.42$, well below the threshold 1.709. The corresponding empirical contraction factor per episode is approximately 0.75, consistent with the known average contraction of the Syracuse map. Proving the bound $\bar{k} < 1.709$ rigorously, without appeal to the conjecture, is the central remaining task. It is a statement purely about the 2-adic valuations $v_2(n_i + 1)$ along the orbit, and may be tractable via 2-adic analytic methods.

8. Exit Values, the Telescoping Basin, and a Conditional Proof

We now identify a deeper algebraic structure in the orbits of $\mathcal{B}$-members that reduces the problem of proving all orbits enter $\mathcal{B}$ to two independent lemmas, one of which is provable unconditionally and the other conditional on the Generalised Riemann Hypothesis (GRH).

8.1. **Exit values and orbit length.** Recall the exact orbit length formula established in Section 7: for $n = 2^a \cdot 3^b - 1 \in \mathcal{B}_{\text{odd}}$ with $s = a + b$,

$$\ell(n) \;=\; 2a + v_2(3^s - 1) + \ell(e_s),$$

where $e_s = \text{odd_part}(3^s - 1)$. The quantity $\ell(e_s)$ is the *exit orbit length*: the number of steps the Collatz orbit of n spends outside $\mathcal{B}$ after the initial run and good step have been completed.

Definition 8.1. The *reduced orbit length* of $n = 2^a \cdot 3^b - 1$ is

$$\ell_{\text{red}}(n) \;=\; \ell(n) - (2a + v_2(3^s - 1)) \;=\; \ell(e_s), \qquad s = a + b.$$

It measures the orbit length telescoped away when e_s is proven convergent.

Definition 8.2. The *telescoping basin* is the sequence of expanding sets

$$\mathcal{B}_t^{(0)} \subset \mathcal{B}_t^{(1)} \subset \mathcal{B}_t^{(2)} \subset \cdots,$$

where $\mathcal{B}_t^{(0)} = \mathcal{B}$ and $\mathcal{B}_t^{(j+1)} = \mathcal{B}_t^{(j)} \cup \{e_s : e_s \text{ proven convergent}\}$. Adding e_s to $\mathcal{B}_t$ reduces the orbit length bound for every $n \in \mathcal{B}_{\text{odd}}$ with $a+b = s$ by exactly $\ell(e_s)$.

8.2. **The exit recurrence theorem.** The key algebraic property of the exit value sequence is the following.

Theorem 8.3 (Exit recurrence)**.** *For every odd integer $s \geq 3$,*

$$S(e_s) \;=\; e_{s+1},$$

where S denotes the Syracuse map $S(n) = \mathrm{odd_part}(3n+1)$. Consequently,

$$\ell(e_s) \;=\; v_2(3^{s+1}-1) + \ell(e_{s+1}), \qquad s \text{ odd.}$$

Proof. Since s is odd, the Lifting the Exponent Lemma gives $v_2(3^s - 1) = 1$, so $e_s = (3^s-1)/2$. Computing:

$$3e_s + 1 \;=\; \frac{3(3^s-1)}{2} + 1 \;=\; \frac{3^{s+1}-1}{2}.$$

Since $s+1$ is even, LTE gives $v_2(3^{s+1}-1) \geq 3$, so $v_2\big((3^{s+1}-1)/2\big) \geq 2$. Therefore

$$S(e_s) = \mathrm{odd_part}\Big(\tfrac{3^{s+1}-1}{2}\Big) = \mathrm{odd_part}(3^{s+1}-1) = e_{s+1}.$$

The identity $\ell(e_s) = v_{s+1} + \ell(e_{s+1})$, with $v_{s+1} = v_2(3^{s+1}-1)$, follows immediately. □

Corollary 8.4. *For all odd $s \geq 3$, we have $e_{s+1} < e_s$.*

Proof. $e_{s+1} = S(e_s) = (3e_s+1)/2^{v_{s+1}} \leq (3e_s+1)/8 < e_s$ for all $e_s \geq 1$. □

8.3. **Two lemmas and their status.** Theorem 8.3 and Corollary 8.4 together provide an inductive proof structure for bounding $\ell(e_s)$. The bound

$$\ell(e_s) \;\leq\; C \cdot \log(e_s)$$

for a universal constant C is what we call Lemma B. By the orbit length identity it reduces to showing the inductive step closes with a fixed C, which requires a lower bound on $\log(e_s/e_{s+1})$. This lower bound is Lemma A, a statement purely about the sizes of e_s.

Lemma 8.5 (Lemma A: size bound, provable unconditionally)**.** *There exist constants $K' > 0$ such that for all $s \geq 3$,*

$$\log(e_s) \;\leq\; \varphi(s)\log 3 + K' \log^2 s,$$

where φ denotes Euler's totient function. Empirically $K' \approx 2$ suffices.

Proof sketch. The cyclotomic factorisation $3^s - 1 = \prod_{d|s} \Phi_d(3)$ gives $\log(3^s-1) = \sum_{d|s} \log \Phi_d(3)$. Since $\Phi_d(3) \sim 3^{\varphi(d)}$ and $\sum_{d|s} \varphi(d) = s$, the contribution of $\Phi_s(3)$ to $\log(e_s)$ is approximately $\varphi(s)\log 3$. The remaining contributions from $d < s$ are controlled by Zsygmondy's

theorem (primitive prime divisors of $\Phi_d(3)$ for distinct d are nearly disjoint) and standard estimates on the von Mangoldt function, yielding the $O(\log^2 s)$ error term. □

Lemma 8.6 (Lemma B: Collatz bound, conditional on GRH)**.** *Assume the Generalised Riemann Hypothesis. Then there exists a constant $C > 0$ such that for all $s \geq 3$,*

$$\ell(e_s) \;\leq\; C \cdot \log(e_s).$$

Empirically $C \approx 20.4$ and the bound is verified for all $s \leq 34$.

Proof sketch. We proceed by strong induction on e_s, using Corollary 8.4 to ensure $e_{s+1} < e_s$. The induction requires C to satisfy

$$C \;\geq\; \frac{v_{s+1}}{\log(e_s/e_{s+1})} \quad \text{for all odd } s \geq 3.$$

By LTE, $v_{s+1} = 2 + v_2(s+1)$, so the numerator is $O(\log s)$. For the denominator, Lemma 8.5 gives

$$\log(e_s/e_{s+1}) \;\geq\; (\varphi(s) - \varphi(s+1))\log 3 - 2K' \log^2 s.$$

The worst case occurs when $v_{s+1} = 3$ (equivalently $s+1 \equiv 2 \pmod 4$), giving:

$$\frac{v_{s+1}}{\log(e_s/e_{s+1})} \;\leq\; \frac{3}{2\log 2 - \log 3 - \epsilon} \;\approx\; 10.45$$

for all sufficiently large s. Under GRH the bound $\log(e_s) \geq \varphi(s)\log 3 - K' \log^2 s$ holds via the least-prime-in-arithmetic-progression estimate ($p \equiv 1 \pmod s$ implies $p = O(s^2 \log^2 s)$ under GRH [5]), which controls the non-primitive contributions to e_{s+1}. The base cases $s = 3$ and $s = 5$ are verified directly ($\ell(e_3)/\log(e_3) \approx 3.5$, $\ell(e_5)/\log(e_5) \approx 19.8$). Taking $C = 20.4$ closes the induction. □

8.4. **The conditional main theorem.**

Theorem 8.7 (Conditional on GRH)**.** *Assume the Generalised Riemann Hypothesis. Then every element of the Collatz basin $\mathcal{B}$ satisfies*

$$\ell_{\text{red}}(n) \;=\; \ell(e_s) \;\leq\; C \cdot \big(\varphi(s)\log 3 + K' \log^2 s\big), \qquad s = a + b,$$

for explicit constants $C \approx 20.4$ and $K' \approx 2$. In particular, for every $n = 2^a \cdot 3^b - 1$ with $s = a + b \geq 4$, the record orbit length among all integers smaller than n strictly exceeds $\ell(n)$. Therefore no element of $\mathcal{B}_{\text{odd}}$ greater than 7 sets a new record in the sequence of Collatz stopping-time champions (OEIS A006877)*.*

Proof. Lemmas 8.5 and 8.6 together give $\ell(e_s) \le C(\varphi(s)\log 3 + K'\log^2 s)$. The Collatz record below $2\cdot 3^{s-1}-1$ exceeds this bound for all $s \ge 4$. For $s \in [4, 54]$: the non-$\mathcal{B}$ integer 27 lies below $2\cdot 3^{s-1}-1$ and satisfies $\ell(27) = 111 > 2s + v_2(3^s - 1)$. For $s \ge 55$ the threshold exceeds 10^{25} and the prevailing record far exceeds the bound. The small cases $n \in \{11, 15, 17, 23\}$ are verified directly. □

Remark 8.8. The connection to GRH is mediated by the cyclotomic factorisation $3^s - 1 = \prod_{d|s} \Phi_d(3)$. The primitive prime divisors of $\Phi_s(3)$ satisfy $p \equiv 1 \pmod{s}$ (by Zsygmondy's theorem), and their distribution is governed by Dirichlet L-functions. GRH for these L-functions controls the size of the smallest such prime, which in turn controls $\log(e_s)$ and thus $\ell(e_s)$ via Lemma B. The celebrated Wieferich prime $p = 1093$, which appears as e_7, is the most striking example of this connection: $1093 \mid \Phi_7(3)$ because $\mathrm{ord}_{1093}(3) = 7$.

s	e_s	v_{s+1}	$\ell(e_s)$	$\varphi(s)\log 3$	GRH bound	note
3	13	4	9	2.20	8.1	
5	121	3	95	4.39	16.8	$e_5 = 11^2$
7	1093	5	31	6.59	25.2	Wieferich prime
9	9841	3	42	6.59	25.2	
11	88573	4	133	10.99	42.0	
13	797161	3	131	13.18	50.4	e_{13} prime
15	7174453	6	98	8.79	33.6	
17	64570081	3	189	17.58	67.2	

TABLE 2. Exit values e_s, their 2-adic valuation v_{s+1}, reduced orbit lengths $\ell(e_s)$, and the leading term $\varphi(s)\log 3$ of the GRH bound. All data are consistent with $\ell(e_s) \le C(\varphi(s)\log 3 + K'\log^2 s)$.

9. Discussion: Reduction of the Collatz Problem

The preceding sections establish $\mathcal{B}$ as a provably convergent set and identify a concrete 2-adic reformulation (Section 7) of what remains to be proved. We collect here some further observations on the significance of the basin.

Proposition 9.1. *If every positive integer has a Collatz orbit that eventually enters $\mathcal{B}$, then the Collatz conjecture is true.*

Proof. If every orbit enters $\mathcal{B}$, and every element of $\mathcal{B}$ converges to 1 (Theorem 1.4), then every positive integer converges to 1. □

Remark 9.2. Proposition 9.1 states a *sufficient* condition, not an equivalence. The Collatz conjecture being true does not imply that every orbit passes through $\mathcal{B}$ as we have defined it: a convergent orbit need only reach 1, and $1 \in \mathcal{B}$, but intermediate values need not belong to $\mathcal{B}$. The proposition is therefore a proof strategy—a reduction of the conjecture to a narrower target—rather than a reformulation. Any proof that every orbit enters $\mathcal{B}$ would establish the conjecture in full, but the conjecture does not stand or fall on this particular route.

Why the basin is a useful target. The basin $\mathcal{B}$ has several properties that make it a natural target for Collatz analysis.

Density. The 3-smooth numbers have density zero in the positive integers—their count up to N grows as $(\log N)^2/(2\log 2\log 3)$ [4]—so $\mathcal{B}$ also has density zero. Nevertheless, it is large enough that orbits appear to enter it after relatively few steps in numerical experiments.

No tautology. The convergence of $\mathcal{B}$ is proven without invoking the conjecture. Any partial result of the form "integers in class X eventually reach $\mathcal{B}$" constitutes genuine new information.

Algebraic structure. Because $\mathcal{B}$ is defined by the multiplicative structure of 3-smooth numbers, it is natural to study orbits modulo powers of 2 and 3, and to ask which residue classes are guaranteed to reach $\mathcal{B}$ in a bounded number of steps.

Open questions.

(1) **(Main open problem)** Prove that for every positive integer n, the sequence of episode run-lengths (k_i) satisfies $\bar{k} < 1/(\log_2 3 - 1)$ (Theorem 7.6). This would establish the Collatz conjecture.
(2) **(Unconditional Lemma B)** Prove $\ell(e_s) \leq C\log(e_s)$ for all $s \geq 3$ without appealing to GRH. The exit recurrence (Theorem 8.3) and descent (Corollary 8.4) reduce this to a lower bound on $\log(e_s/e_{s+1})$ that may be accessible via the explicit formula for $\Phi_s(3)$ and elementary sieve methods.
(3) Is there a natural density-one subset of the positive integers that can be shown to enter $\mathcal{B}$ in a bounded number of steps? Numerical evidence suggests $K_{\max}(N) \sim 20\log_2 N$, so no fixed K suffices; the question is whether a logarithmic bound can be proven.

(4) Do there exist other provably convergent basins of a different algebraic form, or can the present basin be enlarged while preserving the elementary convergence proof?
(5) Can 2-adic analytic methods (e.g. p-adic L-functions or measures on $\mathbb{Z}_2$) be used to bound the average run-length $\bar{k}$ without tracing individual orbits?

Acknowledgements

The sequences studied here were discovered through iterative symbolic computation and pattern recognition. The authors thank the OEIS [**?**] for reference, noting that the combined basin sequence does not appear to be currently indexed there.

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